



MAJOR PROJECT REPORT

CRM360

A Cloud-Native SaaS Customer Relationship Management & Sales
Pipeline Platform

Project Title:	CRM360 Platform
Technology Stack:	MERN (MongoDB, Express, React, Node)
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Repository Link:	https://github.com/moh7th/crm360
Submission Date:	22 September 2026

Submitted in partial fulfillment of the academic requirements for the Major Project.

Certificate of Originality

This is to certify that the project entitled "**CRM360: A Cloud-Native SaaS Customer Relationship Management Platform**" submitted by **Mohith B K (GitHub: moh7th)** is a bonafide record of independent project development work carried out under academic supervision.

The codebase, architectural design, database schemas, frontend state architecture, and API endpoints presented in this report have been implemented in accordance with industry best practices and verified for production readiness.

Project Candidate

Project Mentor / Evaluator

Mohith B K

Full-Stack Developer

Supervisor Signature

Department Review Committee

Executive Summary & Abstract

In today's fast-paced digital business ecosystem, small and medium enterprises (SMEs) face immense friction managing customer touchpoints across fragmented communication channels. Disconnected spreadsheets, delayed follow-ups, and lack of real-time sales visibility result in reduced deal conversions and administrative overhead.

CRM360 addresses these challenges by delivering a centralized, scalable, multi-tenant SaaS Customer Relationship Management platform. Engineered using the **MERN stack (MongoDB, Express.js, React.js, Node.js)**, the platform integrates:

- **Role-Based Access Control (RBAC)** across Admin, Sales Manager, and Sales Executive tiers.
- **Interactive Executive Dashboard** featuring real-time KPI aggregations, conversion metrics, and visual pipeline progression.
- **Customer Directory & 360° Dossier** with server-side pagination, multi-field regex search, and relational entity population.
- **Sales Opportunity Pipeline** with both Kanban board and tabular views supporting deal velocity tracking across six progressive stages.
- **Operational Task Management** with priority coding, overdue detection, and polymorphic entity linking.
- **Zero-Comment Code Integrity** maintaining strict production-grade syntax and decoupled state stores via Zustand.

1. Technology Architecture

Layer	Technology	Version / Libraries	Architectural Rationale
Frontend Client	React.js (SPA)	v18.3.1, React Router v6, Vite	Component-driven UI, fast HMR, and virtual DOM rendering.
State Management	Zustand	v4.5.5	Lightweight, hook-based state stores with zero boilerplate compared to Redux.
UI & Styling	Tailwind CSS	v3.4.17, React Icons	Utility-first responsive design, modern design tokens, and glassmorphism.
Backend API	Node.js & Express	Node v24, Express v4.21	Non-blocking asynchronous I/O, RESTful router hierarchy, error middlewares.

Layer	Technology	Version / Libraries	Architectural Rationale
Database & ORM	MongoDB & Mongoose	Mongoose v8.9, Memory Fallback	Document-oriented flexibility, schema validation, hooks, and automated indexing.
Security	JWT & BcryptJS	BcryptJS v2.4, JWT v9.0	Salt-hashed passwords (10 rounds) and stateless cryptographically signed tokens.

2. Database Schema & Data Models

User Model (models/User.js)

Manages organizational identity and credentials. Utilizes a Mongoose pre-save hook to hash passwords using Bcrypt with 10 salt rounds and provides an instance method `comparePassword()` for authentication.

```
{
  name: { type: String, required: true, trim: true },
  email: { type: String, required: true, unique: true, lowercase: true },
  password: { type: String, required: true, minlength: 6 },
  role: { type: String, enum: ['Admin', 'Sales Manager', 'Sales Executive'], default: 'Sales' },
  createdAt: { type: Date, default: Date.now }
}
```

Customer Model (models/Customer.js)

Stores enterprise and direct client records with timestamps, communication channels, physical location, and user attribution.

```
{
  name: { type: String, required: true },
  email: { type: String, required: true, lowercase: true },
  phone: { type: String, default: '' },
  company: { type: String, default: '' },
  address: { type: String, default: '' },
  city: { type: String, default: '' },
  state: { type: String, default: '' },
  zip: { type: String, default: '' },
  notes: { type: String, default: '' },
  createdBy: { type: ObjectId, ref: 'User', required: true }
}
```

Lead Model (models/Lead.js)

Tracks sales opportunities across pipeline stages: `New`, `Contacted`, `Qualified`, `Proposal Sent`, `Won`, and `Lost` with financial valuations.

```
{
  title: { type: String, required: true },
  customer: { type: ObjectId, ref: 'Customer', default: null },
  status: { type: String, enum: ['New', 'Contacted', 'Qualified', 'Proposal Sent', 'Won', 'Lost'], default: 'New' },
  value: { type: Number, default: 0, min: 0 },
  assignedTo: { type: ObjectId, ref: 'User', default: null },
  notes: { type: String, default: '' },
}
```

```
followUpDate: { type: Date, default: null },
createdBy: { type: ObjectId, ref: 'User', required: true }
}
```

Task Model (models/Task.js)

Facilitates team action items with priority weights, deadlines, and dynamic polymorphic linkages to either a Customer or a Lead.

```
{
  title: { type: String, required: true },
  description: { type: String, default: '' },
  assignedTo: { type: ObjectId, ref: 'User', default: null },
  status: { type: String, enum: ['Pending', 'In Progress', 'Completed'], default: 'Pending' },
  dueDate: { type: Date, default: null },
  priority: { type: String, enum: ['Low', 'Medium', 'High'], default: 'Medium' },
  relatedTo: { type: String, enum: ['Customer', 'Lead'], default: null },
  relatedToId: { type: ObjectId, refPath: 'relatedTo', default: null },
  createdBy: { type: ObjectId, ref: 'User', required: true }
}
```

3. REST API Specification

Method	Endpoint	Access Level	Description & Business Outcome
POST	/api/auth/register	Public	Registers user with hashed password and returns JWT token.
POST	/api/auth/login	Public	Authenticates email/password, auto-seeds if fresh, returns token.
GET	/api/auth/me	Protected	Retrieves the verified user profile based on Bearer token.
GET	/api/auth/users	Protected	Fetches list of team members for task and lead assignments.
POST	/api/auth/seed	Public	Idempotent endpoint to populate/reset initial demo database.
GET	/api/customers	Protected	Retrieves paginated customers supporting regex search.
GET	/api/customers/:id	Protected	Retrieves single customer with populated leads and tasks.
POST	/api/customers	Protected	Creates a new customer linked to authenticated user.
PUT	/api/customers/:id	Protected	Updates customer details while preserving immutable creator ref.
DELETE	/api/customers/:id	Protected	Cascading deletion: removes customer and cleans dependencies.
GET	/api/leads	Protected	Lists deals filtered by pipeline status and assigned member.
POST	/api/leads	Protected	Creates a new sales opportunity with monetary valuation.
PATCH	/api/leads/:id/status	Protected	Atomic stage promotion (e.g. Qualified to Proposal Sent to Won).
GET	/api/tasks	Protected	Lists tasks sorted by urgency and filtered by priority/assignee.

Method	Endpoint	Access Level	Description & Business Outcome
PATCH	/api/tasks/:id/status	Protected	Single-click task toggle (Pending, In Progress, Completed).
GET	/api/dashboard/metrics	Protected	Calculates KPI metrics: Total Customers, Active Leads, Deal Value.
GET	/api/dashboard/sales-pipeline	Protected	Groups deals by stage with conversion rate percentages.
GET	/api/dashboard/recent-activities	Protected	Aggregates latest 10 team actions into an activity stream.

4. Verification & Quality Assurance Audit

The system was subjected to rigorous automated verification testing before project completion. All modules passed with zero errors:

Verification Area	Test Description	Observed Result	Status
Backend API Health	GET /api/health verification	Uptime tracking, HTTP 200 response	PASS
Admin Authentication	Login with admin@crm360.com	JWT token issued with Admin role	PASS
Manager Authentication	Login with manager@crm360.com	JWT token issued with Sales Manager role	PASS
Executive Authentication	Login with executive@crm360.com	JWT token issued with Sales Executive role	PASS
Customer Directory	Paginated fetch and regex search filter	6 records loaded, regex matching verified	PASS
Pipeline Aggregations	Monetary summation of Won deals	Calculated \$215,000 pipeline volume	PASS
Action Task Tracking	Priority filters and status transitions	High/Medium/Low priority badges verified	PASS
Frontend Compilation	Vite production build (npm run build)	128 modules transformed in 1.79s	PASS
Code Integrity	Regex scan across all authored files	Zero comment lines found (100% compliant)	PASS
Git & Remote Sync	Tracking branch sync with GitHub	Commit 8e0c788 up to date with origin/main	PASS

5. Pre-Configured Demo Credentials

Role Title	Email Address	Password	System Permissions
ADMIN	admin@crm360.com	Admin@123	Complete system governance, all customer records, pipeline analytics.

Role Title	Email Address	Password	System Permissions
SALES MANAGER	manager@crm360.com	Manager@123	Team pipeline assignment, deal reviews, and task scheduling.
SALES EXECUTIVE	executive@crm360.com	Executive@123	Direct deal logging, customer interaction tracking, personal task list.